

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	G.Hima Bindu
Register Number	192511377

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I G.Hima Bindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *G. Hima Bindu*

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Introduction:

Many companies sell their products through online shopping websites. During festival sales like Amazon Great Indian Festival or Flipkart Big Billion Days, millions of customers visit the website at the same time. If the website has only a few servers, it may become slow or stop working. Cloud computing solves this problem by using Auto-Scaling and Load Balancing. These services help the website work smoothly even during heavy traffic.

Answer

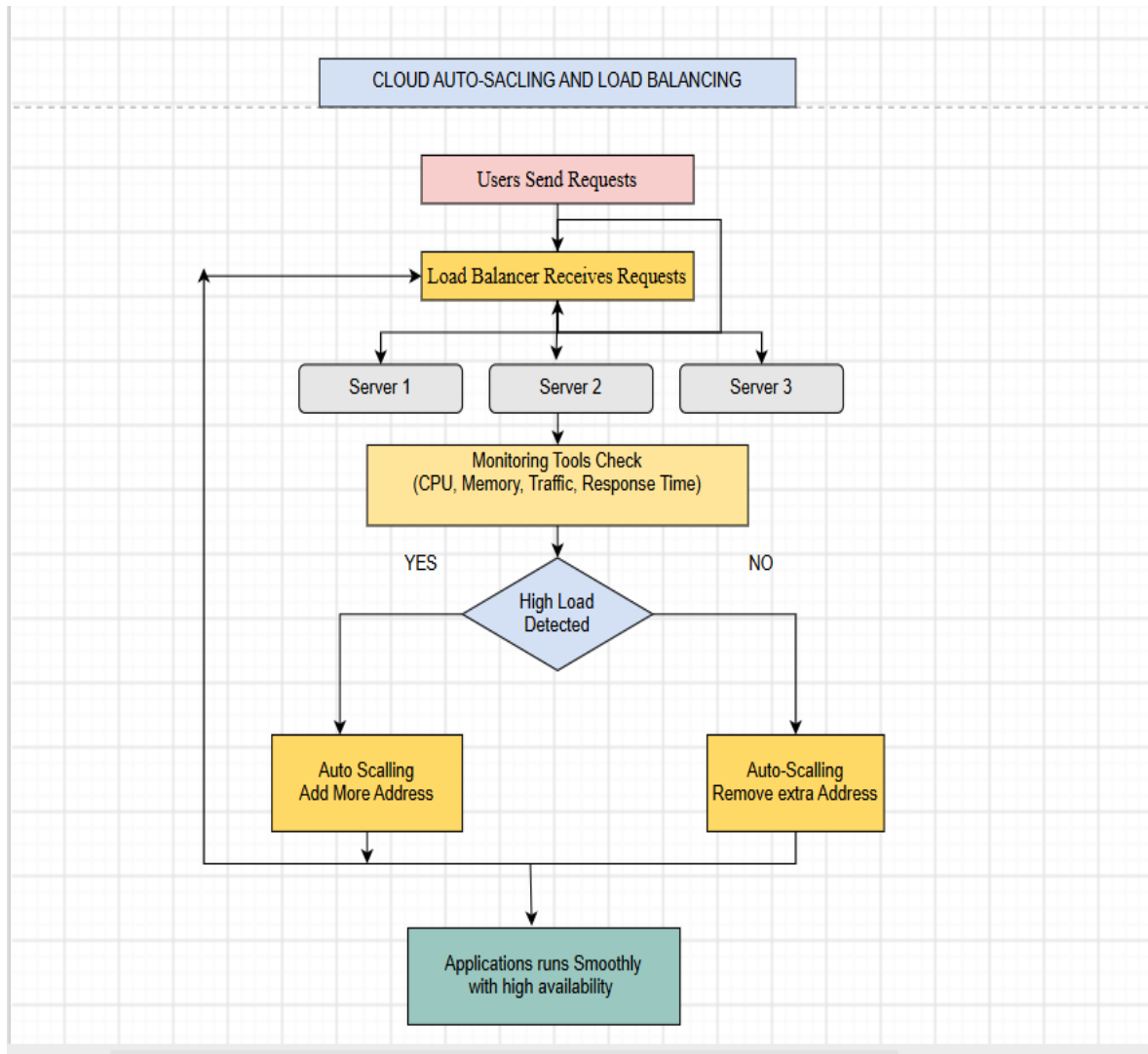
Auto-Scaling is a cloud feature that automatically increases or decreases the number of servers depending on the number of users. When many customers visit the website, the cloud starts additional servers automatically. When the traffic becomes low, the extra servers are removed. This helps the company save money because it only pays for the resources it uses.

Load Balancing is another important cloud service. It receives all the user requests and distributes them equally among different servers. This prevents one server from becoming overloaded. If one server fails, the load balancer automatically sends the requests to another working server. Because of this, the website continues to work without interruption.

These two services together reduce downtime and improve the user experience. Customers can open web pages quickly, search for products without delay, and complete payments smoothly. They do not face website crashes even during heavy traffic.

Monitoring tools continuously check CPU usage, memory usage, network traffic, and server health. When the system detects high resource usage, it automatically starts more servers through Auto-Scaling. Monitoring tools also send alerts if any server has a problem.

A good example is the Amazon Great Indian Festival. During the sale, millions of customers visit the website. Auto-Scaling creates more servers, Load Balancing shares the traffic, and Monitoring tools continuously check the system. Because of this, customers can shop without any interruption and the company can handle large numbers of orders successfully.



2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Introduction

Every organization stores important business data such as customer information, financial records, and project files. Losing this data can cause serious problems. Cloud storage is a safe and reliable solution for storing backup data because it provides high security, durability, and easy recovery.

Answer

Cloud storage stores data in multiple locations instead of keeping it in only one place. This is called data redundancy. If one storage server fails because of hardware damage or a natural disaster, another copy of the data is available. This improves the durability of the stored data and reduces the risk of data loss.

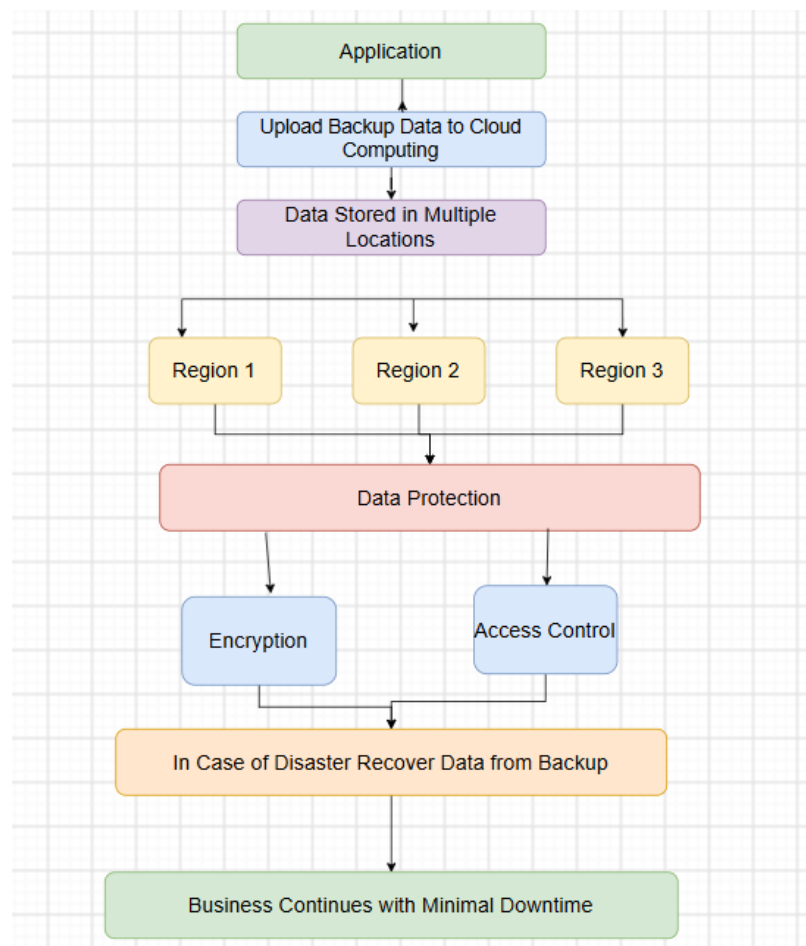
Cloud providers also protect backup data using encryption. Encryption converts the original data into a secure format that cannot be read by unauthorized people. Data is encrypted while it is stored in the cloud and also while it is being transferred over the internet.

Access control provides another layer of security. Only authorized users are allowed to access the backup data using usernames, passwords, Identity and Access Management (IAM), and Multi-Factor Authentication (MFA). This prevents unauthorized access to sensitive information.

Cloud storage is more economical than traditional storage systems. Companies do not need to purchase expensive storage hardware or spend money on maintenance. They only pay for the amount of storage they actually use. Cloud providers also manage backups automatically, reducing administrative work.

A good example is disaster recovery. If a company's office is damaged by fire or flooding, all important files can be restored from cloud backups. Employees can continue their work with very little downtime because the backup data is already stored safely in the cloud.

2.CLOUD STORAGE FOR BACKUP DATA



3.A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Introduction

Many mobile applications are used by people from different countries. If all users access one central server, the application may become slow. A Content Delivery Network (CDN) helps solve this problem by delivering content from servers located close to the users.

Answer

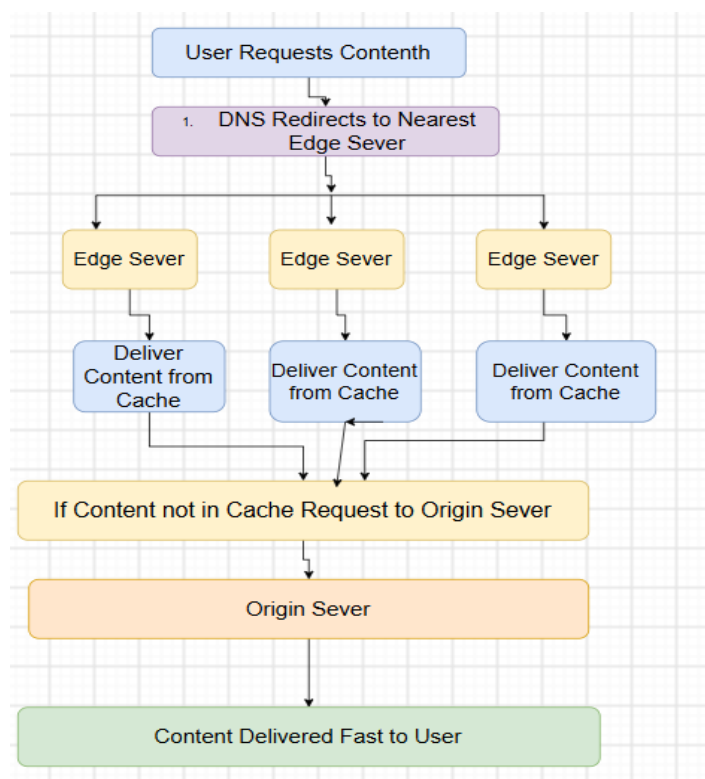
A Content Delivery Network is a group of servers placed in different parts of the world. These servers store copies of frequently used content such as images, videos, and web pages. When a user opens the application, the content is delivered from the nearest server instead of the main server.

Edge servers reduce the distance between users and the cloud server. Because of this, data travels faster and users receive content with less delay. This reduces latency and improves the overall performance of the application.

Cloud providers have data centers in many countries. If one data center becomes unavailable, another data center continues providing the service. This improves global availability and keeps the application running without interruption.

A good example is Netflix. When users watch movies, the videos are delivered from the nearest edge server. This reduces buffering and provides smooth video streaming. Online gaming companies also use CDNs to reduce delays and improve the gaming experience for players around the world.

3.CONTENT DELIVERY NETWORK(CDN)



4.A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Introduction

Banks and financial institutions store highly confidential customer information. They must protect this information from cyber attacks and also follow government rules and regulations. Cloud providers offer many security features to meet these requirements.

Answer

Cloud providers support security standards such as ISO 27001, PCI-DSS, GDPR, and SOC 2. These standards help organizations follow legal and industry security requirements.

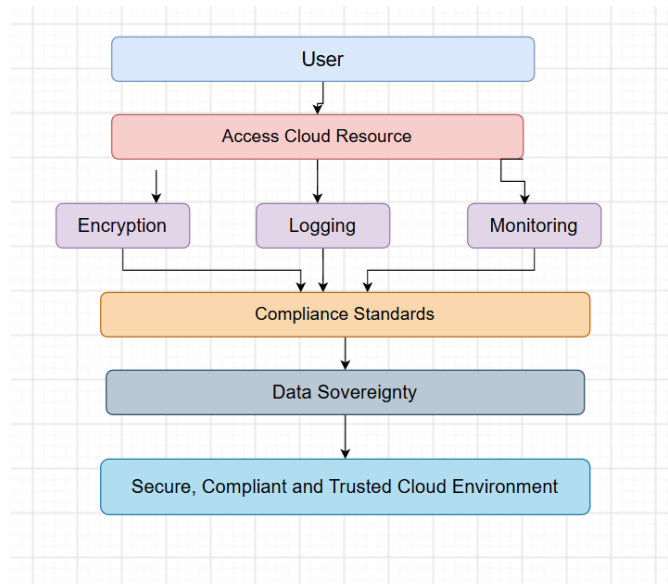
Encryption protects customer information while it is stored in the cloud and while it is transferred through the internet. Even if someone accesses the data without permission, they cannot understand it because it is encrypted.

Logging records every activity performed by users and administrators. Monitoring tools continuously check for suspicious activities and send alerts whenever unusual behavior is detected. These records are useful during security audits and investigations.

Data sovereignty means customer data must be stored within a specific country according to government laws. Cloud providers allow organizations to choose the storage location so that legal requirements are followed.

For example, a bank stores customer transaction records only in Indian cloud data centers. Every transaction is encrypted, user activities are logged, and monitoring tools detect unauthorized access attempts. This helps protect sensitive financial information and maintain customer trust.

4.CLOUD SECURITY AND COMPLIANCE



5.A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Introduction

Modern applications are becoming larger and more complex. To manage them easily, software developers use containers. Containers package the application along with all the files and software needed to run it. Kubernetes is used to manage these containers automatically.

Answer

Containers are lightweight compared to virtual machines because they share the same operating system instead of running separate operating systems. They start quickly, use less memory, and provide better performance. Virtual machines require more storage and system resources because each virtual machine contains its own operating system.

Kubernetes is a container orchestration platform. It automatically deploys containers, manages them, and increases or decreases the number of containers according to user demand. If one container fails, Kubernetes automatically starts another container so that the application continues running without interruption.

One of the biggest advantages of containers is portability. The same container can run on AWS, Microsoft Azure, Google Cloud Platform, private cloud, or hybrid cloud without changing the application. This makes deployment faster and easier.

A good example is an online shopping application. Different services such as User Login, Product Catalog, Shopping Cart, Payment, and Order Management run in separate containers. Kubernetes manages all these containers, balances the workload, and automatically scales the application when more users access the website.

5.Containerization and Kubernetes

